

Solution Requirements

Date	04 July 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements

The HDI Predictor System uses Machine Learning to predict the Human Development Index (HDI) based on four development indicators. The system accepts user inputs, processes them using a trained ML model, and displays the predicted HDI score through a web application.

- **Functional Requirements (FR):** Describe what the system does, such as login, prediction, data validation, and report generation.
- **Non-Functional Requirements (NFR):** Describe how the system performs, such as speed, security, usability, and reliability.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Users can log in securely to access the system.	High
2	Prediction	Predicts HDI from user inputs.	High

S.No	Requirement Category	Requirement Description	Priority
3	Data Validation	Validates input values before prediction.	High
4	User Interface	Displays dashboard and prediction result.	High
5	Reporting	Generates and downloads prediction reports.	Medium
6	Data Storage	Saves prediction history.	Medium
7	Export	Exports results as PDF or JSON.	Medium
8	Country Selection	Loads country data automatically.	Low

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Targetmetric / Acceptance Criteria
1	Performance	Fast HDI prediction.	Response within 2 seconds
2	Security	Secure login and user data.	Authorized access

S.No	NFR Category	Requirement Description	Targetmetric / Acceptance Criteria
3	Reliability	Accurate and stable predictions.	High prediction accuracy
4	Usability	Simple and responsive interface.	User-friendly design
5	Scalability	Supports future updates.	Easy to extend
6	Maintainability	Easy to modify and maintain.	Modular code structure